

Alpilla 600 MW Combined Cycle Power Plant

Electrical Design Criteria

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1. Purpose and scope

1.1 This document defines the electrical design criteria for the Alpilla 1 x 1 multi-shaft combined cycle power plant (net 600 MW): system voltages and earthing, short-circuit ratings, auxiliary power redundancy, motors, cables, protection, earthing and lightning protection, lighting, hazardous areas, emergency power, and the electrical studies that confirm them. It is the basis for the electrical specifications, calculations, single line diagrams and material requisitions of the project.

1.2 Scope: the 380 kV GIS interface and connection to the grid, generators (10MKA, 20MKA), generator circuit breakers and busducts, generator step-up transformers (10BAT, 20BAT), unit auxiliary transformers (10BBT, 20BBT), the 10.5 kV and 0.42 kV distribution, the emergency diesel generator, UPS and 220 V DC systems, cables and cable routes, earthing and lightning protection, lighting and small power, electrical protection and revenue metering. The generator and GSU internals supplied by the power island supplier (IEC) are covered only where they interface with the balance-of-plant electrical system.

1.3 Out of scope: the I&C design criteria (ALP-EPC-00000-IC-DBR-0001), the cathodic protection design (ALP-EPC-00000-EL-CAL-0005) and the construction power supply (ALP-EPC-00000-EL-SLD-0002); they are referred to where they interface.

1.4 Status of values: values marked preliminary are IEPC design criteria that hold until the electrical load list (ALP-EPC-00000-EL-LST-0001) and the studies of section 16 are issued; a change to a preliminary value is made in this document and reported to the affected specifications. The electrical auxiliary system concept is fixed by DEC-EPCE-0016.

2. Governing requirements

ER-11.01: Connection to the 380 kV transmission system via an indoor GIS switchyard in a ventilated building (salt atmosphere), double busbar and bus coupler, two line bays and two generator transformer bays, designed for 50 kA short-circuit current; outdoor bushings with creepage per ER-02.10.

ER-11.02: Air- or hydrogen-cooled synchronous generators per IEC 60034, 50 Hz, static excitation with PSS, generator circuit breakers per IEC/IEEE 62271-37-013.

ER-11.03: One GSU per generator, ONAN/ONAF, rated for the maximum generator output at -8 degC, on-load tap changer, per IEC 60076; fire walls between transformers.

ER-11.04: Two unit auxiliary transformers, 10.5 kV MV switchboards (2 sections with bus tie), 0.4 kV PMCC with automatic transfer. Start-up power from the grid via back-feed through the GSU.

ER-11.05: Emergency diesel generator for safe shutdown loads (turning gear, lube oil, emergency lighting); the diesel fire pump has its own independent drive; UPS (2x100 %) for DCS and critical loads with 60 min autonomy; 220 V DC and 24 V DC systems.

ER-11.06: Duplicated main protection (Main 1/Main 2 from different manufacturers where required by the TSO) for generators, transformers, busbars and lines.

ER-11.07: Earthing grid per IEEE 80 with safe touch and step voltages; lightning protection per IEC 62305.

ER-11.08: Cables per IEC standards, halogen-free flame-retardant inside buildings, fire-rated cables for safety circuits.

ER-11.09: Revenue metering per TSO/market operator rules (main and check meters, class 0.2S), RTU and telecommunication to the national dispatch centre.

ER-04.07: The plant shall provide primary and secondary frequency control, fast frequency response and reactive power support as required by the Grid Code, including operation within 47.5 Hz to 52.5 Hz for the durations specified therein.

ER-04.08: Each generator shall be capable of power factor 0.85 lagging to 0.95 leading at rated active power at the generator terminals.

ER-04.09: The unit shall survive a full load rejection to house load and remain in stable operation for at least 60 minutes.

ER-04.10: Auxiliary power consumption at base load shall not exceed 2.2 % of gross output; the value is included in the net output guarantee.

ER-02.12: Lightning protection per IEC 62305 with a risk assessment based on 26 thunderstorm days per year and ground flash density 2.4 flashes/km²/year (ALP-OWN-MET-001 section 5).

ER-14.06: Hazardous area classification per IEC 60079-10-1; equipment certified accordingly.

2.1 The requirements above are binding. Where a criterion in this document is stricter than a requirement it is stated as a design margin; where this document appears to conflict with a requirement, the requirement governs and the conflict is reported by a technical query.

3. Applicable site and design data

Parameter	Value	Condition	Status	Basis
Ambient temperature - maximum for full capability	40 degC		confirmed	ER-02.02
Ambient temperature - minimum for full capability	-8 degC		confirmed	ER-02.02
Ambient temperature - survival range	-15 to +45 degC		confirmed	ER-02.02
Annual mean temperature (measured)	14.9 degC	2015-2024 site mast	confirmed	SRC-OWNR-0001
Chloride deposition maximum	420 mg/m ² /day	ISO 9225, category S3	confirmed	SRC-OWNR-0001, ER-02.10
Ground flash density	2.4 flashes/km ² /year	26 thunderstorm days/year	confirmed	SRC-OWNR-0001, ER-02.12
Performance test ambient range	5 to 30 degC		confirmed	SRC-OWNR-0001, ER-02.11
PM10 maximum (event)	260 ug/m ³	Saharan dust episodes	confirmed	SRC-OWNR-0001, ER-02.10
Record maximum / minimum temperature	39.6 / -11.4 degC	1960-2024	confirmed	SRC-OWNR-0001
Sea-salt aerosol annual mean	14 ug/m ³	as NaCl	confirmed	SRC-OWNR-0001
Sea-salt aerosol maximum (event)	165 ug/m ³	as NaCl, lodos storms	confirmed	SRC-OWNR-0001, ER-02.10
SRC ambient dry-bulb temperature	15 degC	Site Reference Conditions	confirmed	ER-02.03
SRC barometric pressure	1,011.5 mbar	Site Reference Conditions	confirmed	ER-02.03
SRC relative humidity	70 %	Site Reference Conditions	confirmed	ER-02.03
Summer design dry bulb (0.4 %)	32.8 degC	coincident RH 48 %	confirmed	SRC-OWNR-0001, ER-02.11
Winter design dry bulb (99.6 %)	-4.8 degC	coincident RH 88 %	confirmed	SRC-OWNR-0001, ER-02.11
380 kV voltage range	360 to 410 kV	342-420 kV for up to 30 min	confirmed	SRC-OWNR-0006
Connection capacity	640 MW export	40 MW import	confirmed	SRC-OWNR-0006
Generator power factor range	0.85 lagging to 0.95 leading -	at rated active power	confirmed	ER-04.08
Grid connection voltage	380 kV	nominal	confirmed	ER-11.01
Grid frequency	50 Hz	operating range 47.5 to 52.5 Hz per Grid Code	confirmed	ER-04.07
Short-circuit level at 380 kV (2029)	12 (min) to 40 (max) kA	future 50 kA (2035)	confirmed	SRC-OWNR-0006
Switchyard short-circuit rating	50 kA		confirmed	ER-11.01
100-year coastal still water level	0.9 m	national datum; +1.40 m with 0.5 m sea-level rise; floor levels >= +1.90 m (ER-02.09); shoreline run-up +4.8 m	confirmed	SRC-OWNR-0002, ER-02.09
Atmospheric corrosivity category	C5 (ISO 12944-2), durability H -		confirmed	ER-02.08
Basic wind speed	30 m/s	10 min mean, 10 m, 50-year return	confirmed	ER-02.05
Design rainfall intensity	110 mm/h	10 min, 25-year return	confirmed	ER-02.05
Distance to heavy-lift port	18 km by sea	400 t quay crane	confirmed	SRC-OWNR-0005
Ground snow load	0.75 kN/m ²		confirmed	ER-02.05
Groundwater level	13.5 m a.s.l.	brackish, chloride 3,000-8,000 mg/l	confirmed	SRC-OWNR-0008
Heavy transport road limit	120 t gross	two bridges on the state road	confirmed	SRC-OWNR-0005, ER-02.14
Height datum	TUDKA-99 orthometric heights (m a.s.l.)		confirmed	SRC-EPCE-0011

Parameter	Value	Condition	Status	Basis
Plant grid origin in UTM zone 35N (E 0 / N 0)	E 541,250.000 m / N 4,532,480.000 m m	TUREF (ITRF96), south-west plot corner	confirmed	SRC-EPCE-0011
Platform level acceptance tolerance	0.15 m	Owner-levelled platform +15.00 m	confirmed	DEC-OWNR-0002
Plot dimensions	400 x 300 m	12.0 ha; usable about 10.6 ha	confirmed	SRC-OWNR-0005, ER-02.01
Rotation plant grid to UTM grid	-0.3281 deg	plant north = true north; grid convergence at site	confirmed	SRC-EPCE-0011
Scale factor plant grid to UTM	0.999612 -	combined UTM point scale and height factor	confirmed	SRC-EPCE-0011
Seismic design PGA (DD-2, 475-year)	0.45 g	TBDY 2018, site-specific hazard study	confirmed	ER-02.06
Seismic site class	ZD (Vs30 = 240 m/s) -	TBDY 2018	confirmed	SRC-OWNR-0008
Site grade level	15 m a.s.l.		confirmed	ER-02.01

3.1 Electrical equipment shall be rated for the ambient and site conditions of the parameters above, including the ambient temperature range for full capability (DP-CASE-0007, DP-CASE-0008), the survival range (DP-CASE-0009) and the C5 marine atmosphere (DP-CASE-0003).

3.2 The 380 kV system operates within the voltage range of DP-OWNR-0050 and with the short-circuit level range of DP-OWNR-0049. The lower value shall be used for generator synchronisation, voltage regulation and minimum-fault protection sensitivity; the upper value for the equipment ratings of the GIS bays, together with the switchyard rating of DP-CASE-0025.

3.3 The 380 kV export shall not exceed the connection capacity of DP-OWNR-0051. The rated connection current of the GIS line bays and the GSU rating shall be checked at the winter design case (WIN-NG-100) with the export limiter of TQ-IEC-007 in action.

3.4 Seismic loads for all electrical equipment, anchorage, cable supports and battery racks follow TBDY 2018 with the PGA of DP-CASE-0002 and site class of DP-OWNR-0048. Equipment shall be qualified by type test or analysis per IEC 60068-3-3 or IEEE 344 for the required response spectrum, and shall keep its function (no trip, no loss of supply) after the design earthquake.

3.5 The groundwater level is only 1.5 m below site grade (DP-OWNR-0047 against DP-CASE-0001). Cable basements, cable trenches and earthing pits below +14.0 m shall be designed as water-tight structures or use water-blocked cables in sealed ducts; no MV cable shall lie in a trench that can flood.

4. Codes, standards and precedence

4.1 Precedence: (1) Turkish law and regulations (Grid Code, Heavy Current Installations Regulation, Earthing Regulation, Explosive Atmospheres Regulation), (2) the Employer's Requirements, (3) IEC standards for design, manufacture and testing, (4) IEEE / NFPA standards where cited or where no IEC standard exists. A stricter requirement always governs.

4.2 Power installations above 1 kV shall comply with IEC 61936-1, low-voltage installations with IEC 60364, short-circuit calculations with IEC 60909, the GIS with IEC 62271-203, generators with IEC 60034-1, generator circuit breakers with IEC/IEEE 62271-37-013, transformers with IEC 60076 and communications with IEC 61850.

4.3 Switchgear and controlgear shall additionally comply with IEC 62271-200 (MV metal-enclosed) and IEC 61439 (LV assemblies), with internal arc classification, and shall be type-tested. Standards named in this document but not yet in the project register are proposed for registration (see findings).

4.4 Design values that come from a design_parameter, equipment or decision record are cited in this document and shall not be re-derived from other text; a change in a record is followed by a review of this document (R-002, R-004).

5. System voltages, frequency and earthing

System	Voltage / earthing	Remarks
380 kV grid and GIS	380 kV nominal (Um 420 kV), 50 Hz, solidly earthed	Range 360-410 kV per DP-OWNR-0050; 47.5-52.5 Hz per ER-04.07 [DP-CASE-0024, DP-CASE-0026, ER-11.01]
Generator 10MKA (GT)	21 kV, high-resistance earthed	Neutral through distribution transformer with secondary resistor; voltage range +/- 5 % [ER-11.02, DEC-EPCE-0016]
Generator 20MKA (ST)	15.75 kV, high-resistance earthed	As GT generator; ground fault current limited to 10 A primary or less [ER-11.02, DEC-EPCE-0016]
Auxiliary MV system	10.5 kV, resistance earthed	Neutral earthing resistor at each UAT star point; earth fault current limited to 300 A for 10 s [ER-11.04, DEC-EPCE-0016]
LV system	0.42 kV (400 V nominal), TN-S, neutral solidly earthed at the transformer	420 V no-load secondary voltage; 3-phase 4-wire, separate PE conductor throughout [ER-11.04, DEC-EPCE-0016, REF-CONS-0077]

System	Voltage / earthing	Remarks
220 V DC system	220 V DC, unearthed (IT)	Insulation monitoring with earth-fault location; voltage window 187 V to 242 V [ER-11.05, DEC-EPCE-0016]
UPS system	230 V AC, 50 Hz, single phase	Voltage +/- 1 %, THD < 3 %; TN-S at the output [ER-11.05, DEC-EPCE-0016]
Control voltage	24 V DC	From redundant DC/DC converters on the 220 V DC system [ER-11.05]

5.1 The generator neutral earthing shall limit the stator ground fault current to 10 A or less, so that the fault current does not damage the stator core, and shall provide 100 % stator earth fault protection. Ground fault detection on generators shall not depend on the GSU vector group alone.

5.2 The 10.5 kV resistance earthing limits the earth fault current to 300 A for 10 s so that motor and cable damage on an earth fault is limited and selective sensitive earth fault protection is possible. A change of the value shall be verified by the protection coordination study.

5.3 The 0.42 kV system voltage is the no-load secondary voltage of the 10.5/0.42 kV transformers (DEC-EPCE-0016). LV motors are rated 400 V (Zone A, +/- 5 % per IEC 60034-1) so that the 420 V no-load voltage does not exceed the motor and equipment rating; the requirement ER-11.04 designation '0.4 kV' is met by this system.

5.4 The LV transformers shall be cast-resin dry-type transformers to IEC 60076-11 installed indoors, vector group Dyn11, impedance to limit the LV short-circuit level to the switchgear rating, off-circuit tap changer +/- 2 x 2.5 %.

5.5 Oil-filled transformers (GSU, UAT) shall have an oil containment and a fire wall or minimum separation per NFPA 850 and ER-11.03; a fluid of high fire point (natural ester per IEC 62770) is permitted if accepted by the transformer manufacturer.

6. Short-circuit ratings

6.1 Short-circuit currents shall be calculated per IEC 60909 for maximum (voltage factor c max, all sources, tie closed) and minimum (c min, minimum grid level of DP-OWNR-0049, one source) conditions, including the contribution of MV and LV motors. Initial symmetrical current, peak current, breaking current and thermal current shall be reported for every bus.

Item	Rating criterion	Remarks
380 kV GIS	Short-circuit rating of DP-CASE-0025, 1 s (3 s for busbar and earthing switches), 100 % DC component design	Upper Owner grid level of DP-OWNR-0049 plus generator infeed shall not exceed 80 % of the rating [DP-CASE-0025, DP-OWNR-0049, ER-11.01]
10.5 kV switchgear	31.5 kA, 1 s (peak 80 kA), 2,000 A busbar	Calculated maximum Ik" with bus tie closed shall not exceed 80 % of the rating [DEC-EPCE-0016, REF-CONS-0073]
0.42 kV switchgear	Rating of the switchboard equal to or above the calculated Ik" (incl. motor contribution) plus 10 %, rounded to the next standard value (50, 65, 80, 100 kA)	Applies to every LV board and MCC; the interrupting rating of every device shall be equal to or above the bus rating [DEC-EPCE-0016, REF-CONS-0077]
220 V DC	Rating above the calculated battery short-circuit current at the bus	Batteries plus chargers; the fuses / breakers shall interrupt at the bus level [DEC-EPCE-0016]
Generator circuit breaker	Symmetrical and asymmetrical breaking capacity for the system-source and generator-source fault per IEC/IEEE 62271-37-013	Includes the DC time constant of the generator [ER-11.02, REF-CONS-0038]

6.2 Where the calculated short-circuit current with the MV bus tie closed (transfer condition) exceeds the switchgear rating, the transfer shall be restricted (open transition) instead of increasing the rating. Parallel operation of the two LV sections A and B shall not be permitted in normal operation; the LV coupler is interlocked and closes only after the opening of the incoming breaker of the section to be supplied.

6.3 Short-circuit limitation by transformer impedance shall be preferred to current-limiting reactors. Unit auxiliary transformers shall have an impedance that keeps the MV level within the 31.5 kA rating for the maximum case of section 16.

7. Auxiliary power system and redundancy philosophy

Id	Consumer	Method	Power kW	Status
AL-01	GT package auxiliaries (lube oil, hydraulic, enclosure ventilation, fuel skid, filter house)	fixed	1,180	preliminary
AL-02	GT generator auxiliaries (seal oil, stator water, excitation)	fixed	420	preliminary
AL-03	ST auxiliaries (lube oil, EHC, gland steam exhausters)	fixed	380	preliminary
AL-04	ST generator auxiliaries (excitation, TEWAC)	fixed	260	preliminary
AL-05	HRSG auxiliaries (CPH recirculation pump, ammonia skid, drains)	fixed	210	preliminary
AL-06	Condenser vacuum pump (1 running)	fixed	75	preliminary
AL-07	Turbine control system and UPS load	fixed	60	preliminary

Id	Consumer	Method	Power kW	Status
AL-08	GT GSU losses at this case load	fixed	1,107	preliminary
AL-09	ST GSU losses at this case load	fixed	584	preliminary
AL-10	Boiler feed pumps 2 x 100 % (1 running)	pump_node	computed	assumption
AL-11	Condensate extraction pumps 2 x 100 % (1 running)	pump_node	computed	assumption
AL-12	Circulating water pumps 2 x 50 %	cw_pump	computed	assumption
AL-13	Closed cooling water pumps	fixed	160	assumption
AL-14	Electro-chlorination	fixed	180	assumption
AL-15	Water treatment (SWRO, demineralisation, condensate polishing)	fixed	250	assumption
AL-16	Compressed air	fixed	180	assumption
AL-17	HVAC and lighting	fixed	750	assumption
AL-18	Ammonia and chemical dosing	fixed	40	assumption
AL-19	Fuel gas receiving station	fixed	60	assumption
AL-20	LDO system (standby on gas)	fixed	30	assumption
AL-21	UAT and MV/LV transformer losses	fixed	240	assumption
AL-22	Miscellaneous (workshop, lighting outdoor, cathodic protection)	fixed	300	assumption

7.1 The auxiliary power consumption at base load shall not exceed the value of ER-04.10 (DP-CASE-0031); it is calculated in the electrical load list from the equipment records and checked against the HMB at every load-list issue.

7.2 Unit auxiliary transformers (UAT): two transformers at 2 x 100 % (10BBT from the GT generator, 20BBT from the ST generator), connected between generator and generator circuit breaker so that the plant can run in house-load operation. Each UAT shall carry the full running plant load with the other out of service; the GT static frequency converter (start-up only) is supplied from 10BBT. The transformer ratings are in the records of 10BBT10 and 20BBT10 (DEC-EPCE-0016).

7.3 With 10BBT out of service the plant runs from 20BBT. A GT start from 20BBT alone is possible only with the start-up auxiliary load limited to 5 MW besides the static frequency converter (short-time loading per IEC 60076-7, DEC-EPCE-0016); the load-flow and motor-starting study of section 16 confirms the limit, which is stated in the operating instructions.

7.4 Start-up power without generator in service shall come from the grid by back-feed through the GSU and the generator busduct (ER-11.04). The load flow study shall confirm that the back-feed voltage drop and the GSU tap position permit the start of the largest MV motor and the SFC.

7.5 MV distribution: the 10.5 kV switchboard has two sections A and B with bus tie (00BBA10 / 00BBA20), each section fed from one UAT. In normal operation the bus tie is open and each section carries its own loads (1 x 100 % of duplicated equipment on each section, e.g. BFP A on section A, BFP B on section B). The bus tie closes automatically on loss of one incomer.

7.6 Transfer between sources shall be fast (open-transition, synchro-check, target total interruption of 100 ms or less) for planned and fault-initiated transfers, with residual-voltage supervision for slower transfers, and with load shedding of non-essential motors before re-closing. A transfer that would create a bus voltage below 0.75 pu at the moment of closing is not permitted.

7.7 LV distribution is in four pairs of 2 x 100 % transformers on different MV sections (DEC-EPCE-0016): GT and HRSG 00BFT10/20, ST, condenser and cooling 00BFT70/80, BOP, fuel and buildings 00BFT30/40, and CW intake and pump house 00BFT50/60 (10.5/0.42 kV). Each pair feeds a main switchboard with two sections (00BFA10/20, 00BFA11/21, 00BFA12/22; MCC 00BFA40 at the CW intake), each section with its own incomer and a normally open coupler with the interlock of section 6. Redundant process equipment (2 x 100 %, 2 x 50 % + spare) shall be assigned to different sections and different MV sections so that a single busbar or transformer failure never removes both.

7.8 The load of every LV transformer from the electrical load list, with the other transformer of its pair out of service, shall not exceed 80 % of its AN rating in any heat balance case (DEC-EPCE-0016); the remaining 20 % covers the 10 % load-list margin and design growth. A spare capacity of 20 % shall be provided in the number of feeders of every board.

7.9 LV MCCs 00BFA30 and 00BFA50 to 00BFA80 are double-ended: two bus sections with a tie, fed from the A and B sections of their main switchboard; 00BFA40 is the main board of the CW intake, fed directly from 00BFT50/60. Duplicated consumers are connected to different sections, so that the loss of one section never removes both units of a redundant pair.

7.10 Full load rejection to house load: the aux system shall be supplied from the generator via the UAT and remain in stable operation for at least 60 minutes (ER-04.09). The UAT taps, MV bus voltage and frequency (47.5 to 52.5 Hz) shall be checked in the load rejection case of the stability study; no protective trip of the aux system shall occur for the frequency and voltage swing of the case.

7.11 Electrical equipment shall operate continuously at 47.5 to 52.5 Hz (ER-04.07) and with a voltage/frequency ratio up to 1.05 pu (transformers, motors, drives, converters), and shall ride through a voltage dip to 0.7 pu for 1 s without trip of process-critical drives where the process allows.

8. Motors: voltage level and starting

8.1 Motors of 200 kW and above shall be supplied at 10.5 kV, smaller motors at 0.42 kV (400 V motors). The threshold of DEC-EPCE-0014 is the design criterion for the electrical load list; a deviation for a particular motor (e.g. a 190 kW motor with high starting duty) requires a decision.

8.2 Motor ratings shall be the next standard IEC size above the mechanical duty plus margin of DEC-EPCE-0014 (pumps +10 % flow, +5 to +10 % head). Motors shall be IEC 60034-1 duty S1, insulation class F with temperature rise class B, service factor 1.0, efficiency class IE3 or better (IE4 for continuously running LV motors above 75 kW).

8.3 Motor enclosure: TEFC (IP55 minimum indoors, IP56 outdoors), marine grade C5 painting per ISO 12944-2, stainless steel fasteners and nameplates, space heaters for MV motors and motors with a standstill period, drain holes with protection against condensation. Seawater-side motors (CW pump house) shall have IP56 and an anti-corrosion coating suitable for C5-H.

Item	Criterion	Remarks
Starting method: 0.42 kV motors	Direct on line up to 90 kW; soft starter or star-delta for 90 kW to 200 kW where the voltage dip limit is exceeded	Motors with high inertia use soft starters [DEC-EPCE-0014, REF-CONS-0037]
Starting method: 10.5 kV motors	Direct on line; soft starter or reactor only if the DOL dip limits are exceeded	Reduced-voltage starting shall be shown to reach nominal speed within the thermal limit of the motor [DEC-EPCE-0014]
Variable frequency drives	Only where process control or energy savings justify them	Draws on the CW, condensate and feed pumps; the harmonic study shall show THD (V) $\leq 5\%$ at 10.5 kV and 0.42 kV busbars [DEC-EPCE-0014]
Voltage at the motor terminals during start	Not less than 80 % of rated voltage (85 % for MV motors)	Guarantees a minimum accelerating torque of 10 % margin against the load torque curve [REF-CONS-0037]
Voltage dip on the busbar at the start of one motor	10 % or less on the MV bus, 15 % or less on the LV bus (normal supply)	With one UAT or one LV transformer out of service the limits shall still be met (15 % MV, 20 % LV) or the start blocked by interlock [DEC-EPCE-0016]
Start limits	Starting current $6.5 \times I_n$ or less; starting time within stall time minus 10 %	Data sheets of motors shall show the starting torque curve [REF-CONS-0037]

8.4 Motor start shall be checked by the motor starting study for the worst case: minimum grid level of DP-OWNR-0049, maximum load on the bus, one UAT out of service and the emergency diesel generator supply for the safe shutdown loads. Sequenced starting shall be used for groups after a bus transfer (re-acceleration by priority).

8.5 MV motor protection: thermal overload with stall, locked rotor, phase unbalance, undervoltage, earth fault (sensitive), differential protection above 2 MW, RTD and vibration monitoring above 1 MW. Motor protection relays shall communicate on IEC 61850 (section 11).

9. Emergency power, UPS and DC systems

9.1 The emergency diesel generator (00XJA10, 1,250 kVA) supplies the emergency LV switchboard 00BMA10 for the safe shutdown loads: turning gear, lube and jacking oil, seal oil, closed cooling water, UPS and DC chargers, emergency lighting, HVAC of the electrical rooms, control air and fire detection. The load list of the EDG is the safe shutdown load list of the equipment records.

9.2 EDG start: automatic on loss of both normal supplies of 00BMA10 (undervoltage 0.7 pu for 2 s), first supply within 10 s, sequential loading by priority. The largest single step shall not cause a voltage dip above 15 % or a frequency dip above 10 % (ISO 8528-5 class G3), and the loading shall not exceed 80 % of the standby rating. Test start weekly with no load and monthly on load without interruption of supply.

9.3 The EDG shall be rated for the standby (ESP) rating at the site ambient (DP-CASE-0008, 40 degC) and altitude, and have a day tank and fuel storage for at least 8 hours at rated load, refilled from the LDO system. The EDG shall be in an enclosure or room in the marine atmosphere with filtered combustion air (ER-02.10 salt loads).

9.4 The diesel fire pump has its own independent drive; the electric fire pump (LV, DEC-EPCE-0016) is supplied by a dedicated feeder with fire-rated cable and a power transfer to the emergency board. Protection of the fire pump feeders is only short circuit (no thermal overload trip) per NFPA 20. The fire pump controllers shall not depend on the DCS.

9.5 UPS system 230 V AC: two 100 % units (00BRA10) with static bypass. Each UPS shall carry the full UPS load (turbine control, DCS, SIS, CEMS, telecom) at 0.9 power factor and with a 15 % margin, with autonomy of 60 minutes at rated load (ER-11.05). The UPS input is from the LV main board, the DC input from the 220 V DC system; both units feed separate distribution boards with automatic changeover of the loads that have single supplies.

9.6 220 V DC system: two batteries 2 x 100 % with two chargers (00BTA10), two DC bus sections (00BRA20). Battery sizing (ALP-EPC-00000-EL-CAL-0002) shall provide 2 hours of autonomy for the continuous DC load and 30 minutes for the emergency DC oil pumps of the GT and ST at the end of the period, with an ageing factor of 1.25, a temperature factor of 1.1 and the lowest electrolyte temperature of the battery room design. The chargers shall supply the continuous load and recharge the battery within 10 hours.

9.7 Batteries: valve-regulated lead-acid (VRLA) or nickel-cadmium with design life of at least 12 years, in separate rooms with a room temperature of 15 to 25 degC and ventilation for hydrogen (below 1 % H₂ at any time, per IEC 60079-10-1 zoning of battery rooms). The 220 V DC voltage window is 187 to 242 V at the loads; the end-of-discharge voltage per cell shall be checked against the 187 V limit.

9.8 Protection relays and breaker trip circuits shall be supplied from the 220 V DC system with the two protection groups (Main 1 / Main 2) on different batteries; the 24 V DC control voltage is derived from redundant DC/DC converters. All DC and UPS distributions have a supply monitoring alarm to the DCS.

9.9 The plant has no black-start capability requirement in the Employer's Requirements; the emergency diesel generator is sized for safe shutdown, not for the start of the GT. Any Owner request for black start shall be treated as a variation.

10. Cables and cable installation

10.1 Cables shall comply with IEC standards: IEC 60502 (power cables up to 30 kV), IEC 60227 / 60228 (conductors), IEC 60332-3 category A (flame retardance), IEC 60754 and IEC 61034 (halogen-free, low smoke) inside buildings, and IEC 60331 for circuit integrity (ER-11.08).

10.2 Conductor material is copper for power, control and instrument cables. Minimum cross sections: power 2.5 mm², control 1.5 mm², instrument 0.75 mm² (twisted screened pairs, overall screen), earthing 25 mm² for equipment bonding conductors. Aluminium shall not be used.

10.3 Insulation: XLPE, 90 degC conductor temperature in continuous operation and 250 degC in short circuit; MV cables 6/10 (12) kV (for 10.5 kV) with copper tape screen; outer sheath of halogen-free flame-retardant material (LSZH) indoors, and UV-, chloride- and hydrocarbon-resistant sheath outdoors, direct burial and in the CW pump house; cables in the ground water zone shall have radial and longitudinal water blocking.

10.4 Current-carrying capacity per IEC 60364-5-52 and IEC 60287 for the actual installation: air temperature 40 degC (DP-CASE-0008) or 45 degC in areas outside HVAC (DP-CASE-0009 survival range), soil temperature 25 degC at 1.0 m depth with a thermal resistivity of 1.5 K m/W, grouping and proximity factors of the actual tray fill. Cables in conduit or duct banks shall be derated for the worst duct.

10.5 Voltage drop: 3 % or less at the steady load and 5 % or less from the LV transformer terminals to the load; 2 % or less on MV feeders at the steady load; during motor start the terminal voltage limit of section 8. The cable sizing shall use the load-list currents (design current plus 10 % margin) and the shortest protection setting, not the rated current of the breaker.

10.6 Short-circuit withstand: adiabatic verification ($k_2 S_2 \geq I_2 t$) with the maximum short-circuit current at the cable sending end and the total clearing time of the backup protection (not the instantaneous trip), including the screen of MV cables for the earth-fault current of the resistance-earthed system (300 A, 10 s) and for the double earth fault.

10.7 Fire performance: cables of safety circuits (fire pumps, emergency lighting, EDG, fire detection and alarm, UPS and DC distribution to the emergency loads, SIS logic power) shall be fire-rated to IEC 60331 for 90 minutes (180 minutes for the electric fire pump feeder, NFPA 20). Fire-rated cables shall not share tray with non-rated cables of other redundancy.

10.8 Segregation: redundant circuits (A/B feeders, duplicate protection, duplicate control) shall run on separate routes with a fire barrier or at least 1 m distance; MV, LV power, control and instrument cables on separate trays with a spacing of 300 mm between power and instrument trays; signal cables cross power cables at 90 degrees only.

10.9 Cable trays and ladders: hot-dip galvanised steel to EN ISO 1461 with C5 duplex coating or stainless steel 316L in the CW pump house and outdoor areas in direct spray; 30 % spare space on trays, 20 % spare cores (at least 2) in multicore cables, 25 % spare capacity in duct banks.

10.10 Cable supports, trays and penetrations shall be designed for the seismic loads of the site (DP-CASE-0002); penetrations through fire walls and floors shall be sealed to the fire resistance of the wall; entries into buildings shall be sealed against gas and water.

10.11 Cable identification per KKS (ER-01.06); the cable schedule is generated from the cable records. The cable sizing calculation (ALP-EPC-00000-EL-CAL-0001) uses the criteria of this section and the electrical load list.

11. Protection, control and metering philosophy

11.1 Electrical protection shall be based on IEC 61850 (Edition 2) with numerical relays and a redundant station bus (PRP or HSR per IEC 62439-3) between the relays and the electrical system gateway of the DCS. Trip signals of main protection shall also be hardwired to the circuit breaker trip coils; a communication failure shall not disable protection.

11.2 Main protection shall be duplicated (Main 1 and Main 2), from different manufacturers where required by the TSO, for the generators, GSU, busbars and 380 kV lines, on separate current transformer cores, voltage transformer windings, DC supplies (section 9) and trip coils.

11.3 Generator protection: differential (87G), stator earth fault 100 % (64G), loss of excitation (40), negative sequence (46), reverse power (32), overexcitation V/Hz (24), overvoltage and undervoltage (59/27), overfrequency and underfrequency (81), inadvertent energisation (50/27), out-of-step (78), rotor earth fault (64R), backup impedance (21) and breaker failure, per IEEE C37.102. Generator and GSU overall differential (87GT) is applied in addition to 87G.

11.4 Transformer protection (GSU and UAT): differential (87T), restricted earth fault, Buchholz / sudden-pressure relay, oil and winding temperature, overcurrent backup with high-set element, and overexcitation for GSU. UAT backup overcurrent shall be coordinated with the MV feeder protection and shall not trip before the MV bus tie on a section fault.

11.5 380 kV GIS protection: busbar differential (87B) with breaker failure (50BF) and bay units; line protection with line differential (87L), distance (21), directional earth fault, synchro-check and auto-reclose as required by the TSO; overall clearing time (fault to arc extinction) of 100 ms or less for the 380 kV system, breaker-failure backup within 250 ms. TSO requirements of the Grid Code prevail.

11.6 MV and LV protection shall be selective: MV feeders with overcurrent, sensitive earth fault and, for motors, the functions of section 8; LV feeders with breakers / fuses selective to the upstream device. Where the selectivity study shows that the switchgear internal-arc clearing time exceeds 100 ms, arc-fault detection (light plus current) with fast tripping shall be used. Switchgear shall have internal-arc classification (IAC AFLR) at the rated short-circuit current and time.

11.7 Protection settings shall be calculated in the protection coordination study (ALP-EPC-00000-EL-STU-0002) and recorded in the relay setting list (ALP-EPC-00000-EL-LST-0003); relay settings, logic and communications shall be under change control after IFC.

11.8 Synchronisation: automatic synchronisers at the generator circuit breakers (with backup manual synch-check) and at the 380 kV bays; the generator AVR with PSS and voltage control shall meet the Grid Code requirements and shall be tested for the reactive capability of ER-04.08 (DP-CASE-0027).

11.9 Revenue metering: main and check meters of class 0.2S at the 380 kV connection point (dedicated instrument transformer cores of class 0.2 / 0.2S, IEC 61869), sealed, with energy data available to the TSO / market operator; RTU and telecommunication to the national dispatch centre (IEC 60870-5-104) with redundant links. Details in ALP-EPC-00000-EL-SPC-0003.

11.10 Time synchronisation of protection relays, disturbance recorders and DCS shall be from the GPS master clocks with a resolution of 1 ms for the sequence of events (ER-12.08). Relays use PTP (IEEE C37.238) or IRIG-B; see the I&C design criteria for the time synchronisation architecture.

11.11 Electrical control and interlock philosophy is in ALP-EPC-00000-EL-RPT-0001: breakers of the 380 kV and 21 / 15.75 kV are controlled from the DCS with hardwired interlocks in the bay controllers; local control at the switchgear has priority through a local / remote selector; MV / LV boards are supervised by the DCS via IEC 61850 (MV) and Modbus TCP (LV smart devices).

12. Earthing and lightning protection

12.1 The earthing system shall be one common grid for all voltage levels, buildings, steel structures and foundations, designed per IEEE 80 and the Turkish Earthing Regulation, for the safe touch and step voltages of the maximum fault current of the 380 kV system (DP-CASE-0025) with the current split factor and the clearing time of the backup protection (0.5 s minimum design fault duration).

12.2 The soil resistivity shall be measured on site by the Wenner method (four-pin, spacings up to 100 m) and evaluated with the geotechnical investigation (ER-02.07) with seasonal correction; a two-layer soil model shall be used. Seawater and shallow groundwater (DP-OWNR-0047) shall be taken into account.

12.3 The earth grid shall meet the tolerable touch and step voltages of IEEE 80 for a 70 kg person and shall keep the ground potential rise transferred to remote earths (telecommunication, fence, pipes) below the tolerable levels; the grid

resistance shall be 0.5 ohm or less (target) and the actual value shall come from the earthing study (ALP-EPC-00000-EL-STU-0003).

12.4 Materials: bare copper conductor of 120 mm² for the main ring and grid meshes, 95 mm² for risers and 50 mm² for equipment connections, copper-clad or stainless steel rods, exothermic welded or compression joints below ground; no galvanised steel in direct contact with soil or seawater. Bonding to rebar of foundations is permitted where designed together with the cathodic protection (ALP-EPC-00000-EL-CAL-0005); isolating joints shall be provided where the cathodic protection requires them.

12.5 The surface layer of crushed stone (100 mm, resistivity 3,000 ohm m or higher) shall be applied around outdoor transformers and in areas where the touch voltage limit needs it; fences shall be bonded to the grid at intervals of 25 m or connected via separate earth ring.

12.6 Neutral earthing, generator neutral cubicles, surge arresters and transformer neutrals connect to the grid with the shortest path; instrument and communication systems use a separate clean earth bus connected at one point to the plant grid; shields are earthed as defined in the I&C design criteria.

12.7 Lightning protection shall follow IEC 62305 (all parts) with a risk assessment (IEC 62305-2) for every structure, based on the Owner data of ER-02.12: 26 thunderstorm days per year and a ground flash density of DP-OWNR-0009. The risk assessment (ALP-EPC-00000-EL-STU-0004) sets the lightning protection level (LPL) of each structure.

12.8 Minimum protection levels: LPL I for the stack (65 m height), the LDO and ammonia storage, the gas receiving station and other structures containing hazardous areas; LPL II for the buildings that contain electronic systems (control, electrical and GIS buildings) unless the risk assessment permits less; the acceptable risk R1 (loss of human life) is 1E-5 per year or less.

12.9 Metal roofs and steel structures shall be used as natural air terminals and down conductors where their thickness and continuity comply with IEC 62305-3; a lightning protection zone concept (LPZ) with surge protective devices of type 1+2 at the building entries and type 2 / 3 at cabinets and sensitive instrument loops (coordinated per IEC 62305-4). Surge arresters at the GIS cable / line entries and at the GSU and generator terminals per the insulation coordination study.

12.10 Stack obstruction lighting shall comply with the civil aviation requirements of Turkey; the lighting circuit is fed from the emergency board.

13. Hazardous areas and explosion protection

13.1 Hazardous areas shall be classified per IEC 60079-10-1 in the hazardous area classification report and drawings (ALP-EPC-00000-HS-HAZ-0008), based on the properties of the fuel gas (DP-OWNR-0024 to DP-OWNR-0032), the LDO, hydrogen for the generators, aqueous ammonia and battery hydrogen. Electrical equipment shall be selected accordingly and certified (ATEX 2014/34/EU, IECEx) for the zone, gas group and temperature class.

13.2 Minimum equipment protection: zone 1 Ex d, Ex e, Ex p or Ex ia / ib; zone 2 Ex n / ec or better; zone 0 Ex ia only. Gas group IIB + H₂ and temperature class T3 as the minimum for all classified areas (covers natural gas with the hydrogen admixture capability of DP-CASE-0021, LDO vapour and ammonia); group IIC in hydrogen areas (generators, hydrogen supply, battery rooms if classified), where T1 is sufficient. The classification report shall state the group and class per area; a lower group is not permitted without a recorded reason.

13.3 Electrical equipment shall be installed, inspected and maintained per IEC 60079-14 and IEC 60079-17, and recorded in the hazardous area electrical equipment register (ALP-EPC-00000-EL-LST-0004). Cable glands, conduit seals and earthing of Ex equipment are part of the certification.

13.4 Electrical rooms, battery rooms and switchgear shall be located outside classified areas; where a room is close to a zone, the room shall be pressurised (overpressure of at least 50 Pa and airlocks) with filtered air taken from a non-hazardous location and with a fire and gas detection interlock; battery rooms shall have a ventilation of the hydrogen released (EN 50272-2) with hydrogen detection (alarm at 10 % LEL, charger trip at 25 % LEL, HV-DBR-0001 section 8).

13.5 Static electricity and earthing of hazardous area equipment shall be per the explosion protection document (ALP-EPC-00000-HS-RPT-0005); tanks, pipe flanges and loading equipment (LDO unloading) shall be bonded and connected to the earth grid.

14. Lighting and small power

Area	Maintained illuminance (lux) and remarks	Basis
Central control room	500 lx, adjustable 100 to 500 lx, UGR 19 or less, Ra 80 or more	EN 12464-1 office / control room class [ER-11.05]
Offices and workshops	500 lx offices, 300 lx workshops	EN 12464-1
MV, LV, battery and electrical rooms	200 lx at floor, 500 lx at switchgear front panels where read	Emergency lighting on the EDG board
Turbine hall, GT / ST deck, boiler house, pump houses	200 lx (150 lx above 10 m height, 300 lx at maintenance areas)	Localised lighting for inspection points
Outdoor process areas and walkways	50 lx general, 20 lx in walkways, 100 lx at valve stations and instrument points	EN 12464-2
Roads, yards, switchyard and transformer areas	10 lx roads, 20 lx transformer and yard areas	Uniformity 0.25 or better
Escape routes and emergency lighting	1 lx minimum on the centre line, 5 lx at fire-fighting and first-aid equipment, 90 min autonomy	EN 1838; supplied from the emergency board (EDG) with self-contained backup for the transfer time [ER-11.05]

14.1 Luminaires shall be LED, with a lifetime of L70B10 at 50,000 hours or more, maintenance factor 0.8, colour temperature 4,000 K indoors and outdoors, marine grade (aluminium with C5 coating, stainless steel 316L or GRP) and IP66 outdoors. Luminaires in classified areas shall be Ex certified per section 13.

14.2 Lighting circuits are supplied from the LV board of the area; every area of an escape route has two supplies (normal and emergency) and a light control that avoids darkness on a supply loss. The lighting calculation (ALP-EPC-00000-EL-CAL-0004) shall demonstrate the illuminances above at the maintenance factor.

14.3 Small power and welding sockets: 230 V sockets with RCD (30 mA) in buildings, 400 V welding / maintenance sockets (63 A) at intervals of 40 m in the plant with earth leakage protection, IP66 and C5 quality outdoors.

15. Equipment environment, enclosures and insulation

15.1 Electrical rooms shall be air-conditioned with N+1 cooling, positive pressure and filtered air (ER-10.06), to a room temperature of 35 degC or less for switchgear and 30 degC or less for UPS, chargers and control equipment, with the outdoor design temperature of DP-CASE-0008.

15.2 The 380 kV GIS shall be indoors in a ventilated building with salt filtration (ER-11.01). Outdoor insulation (bushings, insulators, surge arresters, cable terminations) shall be designed for the very heavy pollution class of IEC 60815 with the salt aerosol and chloride deposition of DP-OWNR-0005 and DP-OWNR-0007 (ER-02.10), with the specific creepage of the class for the highest system voltage; the pollution class and creepage shall be confirmed by the insulation coordination study.

15.3 Enclosure ratings: switchgear rooms IP4X minimum, outdoor cabinets and junction boxes IP66, outdoor motors IP56, transformer terminal boxes IP54 minimum. Outdoor enclosures shall be of stainless steel 316L or coated to ISO 12944-2 C5-H, with sun shield, anti-condensation heaters and drains.

15.4 Standard insulation levels (Um / power frequency / lightning impulse): 420 kV / 1,050 kV (switching) / 1,425 kV; 12 kV / 28 kV / 75 kV for the 10.5 kV system; the generator terminal equipment per IEC 60034 and the IEC busduct data. The levels shall be confirmed by the insulation coordination study.

16. Electrical studies

16.1 The following studies shall be done by the EPC with the inputs and outputs stated, each as an own document of the project register; results shall be issued as design input before the affected specifications and the requisitions are released for order.

16.2 Load flow (ALP-EPC-00000-EL-STU-0001): network model of the grid (min / max short-circuit level, voltage range of DP-OWNR-0050), GSU, UAT, MV and LV systems for all HMB cases; check the bus voltages (0.95 to 1.05 pu), tap settings, transformer loading, cable and busduct loading, power factor at the connection point, and the export limit of DP-OWNR-0051.

16.3 Short circuit per IEC 60909 (same study): maximum and minimum currents for all buses including motor contribution; results shall confirm the ratings of section 6 and the 380 kV level of DP-CASE-0025; the contribution of the generators, the GT SFC and the back-feed conditions shall be shown.

16.4 Motor starting (same study): start of the largest MV and LV motors, group re-acceleration after transfer, and the SFC start, with the dip limits of section 8 and the EDG load steps of section 9.

16.5 Protection coordination and relay settings (ALP-EPC-00000-EL-STU-0002): settings and time-current curves for generator, transformer, MV, LV and 380 kV protection, arc-flash incident energy per IEEE 1584 and labelling of boards, selectivity of LV devices, and verification of the clearing times of section 11 and the cable withstand of section 10.

16.6 Earthing (ALP-EPC-00000-EL-STU-0003), lightning risk assessment (ALP-EPC-00000-EL-STU-0004) and insulation coordination (ALP-EPC-00000-EL-STU-0005): to the criteria of sections 12 and 15, with the surge arrester ratings and locations, the BIL of the equipment, and the lightning protection level of each structure.

16.7 Grid Code compliance and reactive capability (ALP-EPC-00000-EL-STU-0006): steady-state and dynamic simulation of the power island in the grid model of the TSO (frequency response, voltage control, fault ride-through, reactive capability at the generator and at the connection point per ER-04.08, PSS tuning, stability and house-load transition of ER-04.09), used as basis for the Grid Code tests of ER-16.03.

16.8 Harmonics and power quality: the study shall confirm THD (V) of 5 % or less at the 10.5 kV and 0.42 kV busbars with the SFC and the VFDs in operation, and the flicker and unbalance limits of the Grid Code at the connection point.

16.9 Sizing calculations that follow from this document: cable sizing (ALP-EPC-00000-EL-CAL-0001), battery, UPS and DC (ALP-EPC-00000-EL-CAL-0002), EDG and transformers (ALP-EPC-00000-EL-CAL-0003) and lighting (ALP-EPC-00000-EL-CAL-0004). Each shall cite the criteria of this document by section.

17. Equipment covered by this document

Tag	Description	Service	Redundancy	Rating	Design P / T	Material
00BBA10	MV switchgear 10.5 kV, section A	10.5 kV MV switchboard section A (fed from GT UAT 10BBT)	2 x 100 % (sections A/B with bus tie)	2,000 A, 10.5 kV		Steel enclosure, vacuum circuit breakers, C5 coating
00BBA20	MV switchgear 10.5 kV, section B	10.5 kV MV switchboard section B (fed from ST UAT 20BBT)	2 x 100 % (sections A/B with bus tie)	2,000 A, 10.5 kV		Steel enclosure, vacuum circuit breakers, C5 coating
00BFA10	LV main switchboard 0.42 kV - GT and HRSG, A	LV main switchboard GT and HRSG, section A (from transformer 00BFT10)	2 x 100 % (A/B with coupler)	5,000 A, 400 V		Steel enclosure, form 4b, C5 coating
00BFA11	LV main switchboard 0.42 kV - ST and cooling, A	LV main switchboard ST and cooling, section A (from transformer 00BFT70)	2 x 100 % (A/B with coupler)	2,500 A, 400 V		Steel enclosure, form 4b, C5 coating
00BFA12	LV main switchboard 0.42 kV - BOP, fuel and buildings, A	LV main switchboard BOP, fuel and buildings, section A (from transformer 00BFT30)	2 x 100 % (A/B with coupler)	5,000 A, 400 V		Steel enclosure, form 4b, C5 coating
00BFA20	LV main switchboard 0.42 kV - GT and HRSG, B	LV main switchboard GT and HRSG, section B (from transformer 00BFT20)	2 x 100 % (A/B with coupler)	5,000 A, 400 V		Steel enclosure, form 4b, C5 coating
00BFA21	LV main switchboard 0.42 kV - ST and cooling, B	LV main switchboard ST and cooling, section B (from transformer 00BFT80)	2 x 100 % (A/B with coupler)	2,500 A, 400 V		Steel enclosure, form 4b, C5 coating
00BFA22	LV main switchboard 0.42 kV - BOP, fuel and buildings, B	LV main switchboard BOP, fuel and buildings, section B (from transformer 00BFT40)	2 x 100 % (A/B with coupler)	5,000 A, 400 V		Steel enclosure, form 4b, C5 coating
00BFA30	LV MCC - power block	LV MCC power block (GT/ST/HRSG auxiliaries, closed cooling water)	double-ended, 2 sections with tie (fed from 00BFA10/20)	1,600 A, 400 V		Steel enclosure, form 3b/4b, C5 coating
00BFA40	LV MCC - CW pump house	LV MCC CW pump house (screens, electro-chlorination AL-14, sump pumps, lighting)	double-ended, 2 sections with tie (fed from 00BFT50/60 (main board of the CW intake))	1,600 A, 400 V		Steel enclosure, form 3b/4b, C5 coating
00BFA50	LV MCC - water treatment	LV MCC water treatment (SWRO LV drives, demineralisation, polishing, dosing)	double-ended, 2 sections with tie (fed from 00BFA12/22)	1,000 A, 400 V		Steel enclosure, form 3b/4b, C5 coating
00BFA60	LV MCC - BOP utilities	LV MCC BOP utilities (compressed air AL-16, closed cooling, misc AL-22)	double-ended, 2 sections with tie (fed from 00BFA12/22)	1,600 A, 400 V		Steel enclosure, form 3b/4b, C5 coating
00BFA70	LV MCC - fuel gas and LDO	LV MCC fuel gas and LDO (AL-19, AL-20, LDO forwarding 0.42 MW in LDO case)	double-ended, 2 sections with tie (fed from 00BFA12/22)	1,000 A, 400 V		Steel enclosure, form 3b/4b, C5 coating
00BFA80	LV MCC - buildings and HVAC	LV MCC buildings and HVAC (chillers, AHUs, lighting)	double-ended, 2 sections with tie (fed from 00BFA12/22)	1,600 A, 400 V		Steel enclosure, form 3b/4b, C5 coating
00BFT10	LV transformer 10.5/0.42 kV 3.15 MVA - GT and HRSG, A	LV supply GT and HRSG main switchboard 00BFA10	2 x 100 % (with 00BFT20)	3,150 kVA, 10.5 kV		Cast-resin dry type (IEC 60076-11), IP23 enclosure, indoor transformer room

Tag	Description	Service	Redundancy	Rating	Design P / T	Material
00BFT20	LV transformer 10.5/0.42 kV 3.15 MVA - GT and HRSG, B	LV supply GT and HRSG main switchboard 00BFA20	2 x 100 % (with 00BFT10)	3,150 kVA, 10.5 kV		Cast-resin dry type (IEC 60076-11), IP23 enclosure, indoor transformer room
00BFT30	LV transformer 10.5/0.42 kV 3.15 MVA - BOP, fuel and buildings, A	LV supply BOP main switchboard 00BFA12	2 x 100 % (with 00BFT40)	3,150 kVA, 10.5 kV		Cast-resin dry type (IEC 60076-11), IP23 enclosure, indoor transformer room
00BFT40	LV transformer 10.5/0.42 kV 3.15 MVA - BOP, fuel and buildings, B	LV supply BOP main switchboard 00BFA22	2 x 100 % (with 00BFT30)	3,150 kVA, 10.5 kV		Cast-resin dry type (IEC 60076-11), IP23 enclosure, indoor transformer room
00BFT50	LV transformer 10.5/0.42 kV 1.0 MVA - CW intake, A	LV supply CW intake and pump house	2 x 100 % (with 00BFT60)	1,000 kVA, 10.5 kV		Cast-resin dry type (IEC 60076-11), IP23 enclosure, indoor transformer room
00BFT60	LV transformer 10.5/0.42 kV 1.0 MVA - CW intake, B	LV supply CW intake and pump house	2 x 100 % (with 00BFT50)	1,000 kVA, 10.5 kV		Cast-resin dry type (IEC 60076-11), IP23 enclosure, indoor transformer room
00BFT70	LV transformer 10.5/0.42 kV 1.6 MVA - ST and cooling, A	LV supply ST and cooling main switchboard 00BFA11	2 x 100 % (with 00BFT80)	1,600 kVA, 10.5 kV		Cast-resin dry type (IEC 60076-11), IP23 enclosure, indoor transformer room
00BFT80	LV transformer 10.5/0.42 kV 1.6 MVA - ST and cooling, B	LV supply ST and cooling main switchboard 00BFA21	2 x 100 % (with 00BFT70)	1,600 kVA, 10.5 kV		Cast-resin dry type (IEC 60076-11), IP23 enclosure, indoor transformer room
00BMA10	Emergency LV switchboard	Emergency LV switchboard (EDG and normal supply via main board, ATS)	1 x 100 %	2,000 A, 400 V		Steel enclosure, form 4b, C5 coating
00BMA20	Emergency MCC - power block	Emergency MCC power block (turning gear, emergency lube/jacking oil, chargers)	1 x 100 %	1,250 A, 400 V		Steel enclosure, form 3b, C5 coating
00BRA10	UPS 230 V AC (2 x 100 %)	230 V AC UPS (DCS, SIS, turbine control, CEMS, telecom, fire panels)	2 x 100 %	100 kVA, 230 V		Steel cabinets, VRLA battery
00BRA20	DC 220 V distribution board	220 V DC distribution board (protection, control, emergency oil pumps, UPS inverters)	2 x 100 % (two bus sections)	630 A, 220 V		Steel enclosure, C5 coating
00BTA10	Battery and charger 220 V DC (2 x 100 %)	220 V DC battery and charger (VRLA/NiCd, 108 cells)	2 x 100 %	500 Ah, 220 V		Battery: sealed lead acid / NiCd
00XJA10	Emergency diesel generator 1,250 kVA	Emergency diesel generator for safe shutdown (black plant)	1 x 100 %	1,250 kVA, 1,000 kW, 400 V	50 degC	Steel base frame, acoustic canopy C5
00XJA10BB001	Emergency diesel generator day tank	Emergency diesel generator day tank	1 x 100 %	3 m3	0 barg / 65 degC	Carbon steel with bund
10BBT10	GT unit auxiliary transformer 21/10.5 kV 31.5 MVA	GT unit auxiliary transformer 21/10.5 kV (GT generator terminals to MV section A)	2 x 100 % (with 20BBT10)	31,500 kVA, 21 kV		Steel tank, C5 marine coating
20BBT10	ST unit auxiliary transformer 15.75/10.5 kV 16 MVA	ST unit auxiliary transformer 15.75/10.5 kV (ST generator terminals to MV section B)	2 x 100 % (with 10BBT10)	16,000 kVA, 15.75 kV		Steel tank, C5 marine coating

17.1 The equipment above is the electrical equipment of the auxiliary power system, emergency power and earthing systems as recorded in the equipment records at the time of issue; the rated data are those of the records (single source), not of this document.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
Explosive Atmospheres Regulation	Regulation on Protection of Workers from Explosive Atmospheres
TBDY 2018	Turkish Building Earthquake Code (Türkiye Bina Deprem Yönetmeliği)
Grid Code	Electricity Grid Regulation (Elektrik Sebeke Yönetmeliği) - TSO Grid Code
Heavy Current Regulation	Regulation on Electrical Heavy Current Installations (Elektrik Kuvvetli Akım Tesisleri Yönetmeliği)
Earthing Regulation	Regulation on Earthing in Electrical Installations (Elektrik Tesislerinde Topraklamalar Yönetmeliği)
ATEX 2014/34/EU	ATEX Equipment Directive, transposed as the regulation on equipment for potentially explosive atmospheres
IEC 60034-1	Rotating electrical machines - Part 1: Rating and performance
IEC/IEEE 62271-37-013	High-voltage switchgear - Alternating-current generator circuit-breakers
IEEE C37.102	Guide for AC generator protection
IEC 60076	Power transformers (all parts)
IEC 62770	Fluids for electrotechnical applications - Unused natural esters for transformers
NFPA 850	Recommended Practice for Fire Protection for Electric Generating Plants and HVDC Converter Stations
NFPA 20	Standard for the Installation of Stationary Pumps for Fire Protection
IEC 61936-1	Power installations exceeding 1 kV AC - Part 1: Common rules
IEC 62271-203	High-voltage switchgear - Gas-insulated metal-enclosed switchgear for rated voltages above 52 kV

Code	Title
IEC 60909	Short-circuit currents in three-phase AC systems
IEC 61850	Communication networks and systems for power utility automation
IEC 60364	Low-voltage electrical installations (all parts)
IEEE 80	Guide for Safety in AC Substation Grounding
IEC 62305	Protection against lightning (all parts)
IEC 60815	Selection and dimensioning of high-voltage insulators for polluted conditions
IEC 60079-10-1	Explosive atmospheres - Classification of areas - Explosive gas atmospheres
EN ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles
ISO 12944-2	Paints and varnishes - Corrosion protection of steel structures by protective paint systems - Part 2: Classification of environments